

Stabilization of Cascades Using Integral Input-to-State Stability

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Abstract

We analyze nonlinear cascades in which the driven subsystem is integral ISS, and characterize the admissible integral ISS gains for stability. This characterization makes use of the convergence speed of the driving subsystem, and allows a larger class of gain functions when the convergence is faster. We show that our integral ISS gain characterization unifies different approaches in the literature which restrict the nonlinear growth of the driven subsystem and the convergence speed of the driving subsystem.

1 Introduction

Studies on the stabilization of cascade systems have paved the road to major advances in nonlinear control. Among these advances are several constructive design methods such as *backstepping* and *forwarding*, which are based on recursive applications of cascade designs (see *e.g.* Sepulchre *et al.* [7]), and the discovery of structural obstacles to stabilization such as the *peaking phenomenon*; Sussmann and Kokotović [11].

One of the main motivations for the stabilization of cascades came from the linear-nonlinear cascade

$$\dot{x} = f(x, z) \quad (1)$$

$$\dot{z} = Az + Bu \quad (2)$$

resulting from input-output linearization. Because global asymptotic stability (GAS) of the x -subsystem $\dot{x} = f(x, 0)$ is not sufficient to achieve GAS of the whole cascade with z -feedback $u = Kz$, alternative designs which employ x -feedback were developed, such as the *feedback passivation* design of Kokotović and Sussmann [3]. To achieve GAS

by z -feedback, Sontag [8], Seibert and Suarez [6], Mazenc and Praly [4], Janković *et al.* [2], and Panteley and Loria [5] studied general cascades in which the z -subsystem is nonlinear, and derived conditions for the x and z -subsystems that ensure stability of the cascade. Among these results, a particularly useful one is the input-to-state stability (ISS) condition of [9], which states that if the x -subsystem is ISS with input z , and the z -subsystem is GAS, then the cascade is GAS. This result has been widely used for nonlinear designs based on the *normal form* (1)-(2), in which the *zero dynamics* (1) is ISS. Other results such as [4], [2], [5] make less restrictive assumptions than ISS for the x -subsystem, but restrict the z -subsystem to be locally exponentially stable (LES).

The integral version of ISS (iISS), recently introduced in [10], requires that an energy norm of the input be bounded to ensure boundedness of the states. iISS is less restrictive than ISS because, in an iISS system, a bounded input may lead to unbounded solutions if its energy norm is infinite.

In this paper, we analyze the stability of nonlinear cascades in which the x -subsystem is iISS, and the z -subsystem is GAS. The admissible iISS gains for stability are characterized from the speed of convergence of the z -subsystem. When the convergence is fast, the iISS gain function of the x -subsystem is allowed to be “steep” at zero. We show that this trade-off between slower convergence and steeper iISS gain encompasses, and unifies, several results in the literature. In particular, if the x -subsystem is ISS then the slope of its iISS gain function is very gentle at zero, and tolerates every GAS z -subsystem no matter how slow its convergence is. On the other hand, if the convergence is exponential, that is if the z -subsystem is LES, then the cascade is stable for a large class of iISS gains. This class includes all iISS x -subsystems that are affine in the input z . Thus, for systems like (1)-(2), where a control law can be designed to render

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the z -subsystem GAS and LES, the iISS of the x -subsystem ensures GAS of the cascade.

In Section 2 we define the new concepts used in the paper, and present lemmas which are preliminary to our main results. In Section 3 we present our main result, Theorem 1, which characterizes the admissible iISS gains from the speed of convergence of the z -subsystem. We show that several results in the literature are special cases of Theorem 1, including those that restrict the x -subsystem to be ISS (Corollary 1), and those that restrict the z -subsystem to be LES (Corollary 2). We present examples which illustrate that the restrictions of Theorem 1 are tight, and removing them leads to instability of the cascade. In Section 4, we show that Corollary 2 encompasses previous designs in which the growth of the interconnection term is restricted to be linear. Next, we prove that our result does not require linear growth, and derive a sharp growth condition, which, if violated, results in instability. Conclusions are given in Section 5.

2 Definitions and Preliminary Lemmas

In this section we give definitions, and present lemmas that will be used in the rest of the paper. The proofs are omitted due to space limitations.

Definition 1 *We say that the function $\mu(\cdot) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is class- $\mathcal{K}_o$ if it is class- $\mathcal{K}$, and $\mathcal{O}(s)$ near $s = 0$, that is for all $s \in [0, 1]$, $\mu(s) \leq ks$ for some $k > 0$.*

Definition 2 [10] *The system*

$$\dot{x} = f(x, z) \quad (3)$$

is said to be iISS with input z if there exist a class- $\mathcal{K}_\infty$ function $\omega(\cdot)$, a class- $\mathcal{KL}$ function $\beta(\cdot, \cdot)$, and a class- $\mathcal{K}$ iISS gain $\mu(\cdot)$ such that, for all $t \geq 0$,

$$\omega(|x(t)|) \leq \beta(|x(0)|, t) + \int_0^t \mu(|z(\tau)|) d\tau. \quad (4)$$

Lemma 1 below computes an iISS gain $\mu(\cdot)$ from the derivative of a Lyapunov function:

Lemma 1

i) *If there exists a C^1 , positive definite, radially unbounded function $V(x)$ satisfying*

$$\frac{\partial V}{\partial x} f(x, z) \leq -\rho(|x|) + \mu(|z|) \quad (5)$$

for some positive definite function $\rho(\cdot)$, then the system (3) with input z is iISS with gain $\mu(\cdot)$.

ii) *If there exists a C^1 , positive definite, radially unbounded function $V(x)$ satisfying*

$$\frac{\partial V}{\partial x} f(x, z) \leq \sigma(|z|) \quad (6)$$

for some class- $\mathcal{K}$ function $\sigma(\cdot)$, and if the system (3) is GAS when $z \equiv 0$, then it is iISS with gain $\mu(\cdot) = \sigma(\cdot) + \theta_o(\cdot)$, where $\theta_o(\cdot)$ is a class- $\mathcal{K}_o$ function.

The following lemma shows that if (3) is affine in the input z , then $\sigma(\cdot)$ in (6) is class- $\mathcal{K}_o$, which ensures that the iISS gain $\mu(\cdot) = \sigma(\cdot) + \theta_o(\cdot)$ is also class- $\mathcal{K}_o$:

Lemma 2 *If the input-affine system*

$$\dot{x} = f(x) + g(x)z \quad (7)$$

is iISS, then it is also iISS with a class- $\mathcal{K}_o$ gain $\mu(\cdot)$.

It is known [10] that ISS implies iISS. We further show that ISS allows us to select the iISS gain $\mu(\cdot)$ in (4) to match any desired class- $\mathcal{K}$ function $\tilde{\mu}(\cdot)$ locally:

Lemma 3 *Suppose the system (3) is ISS. Then, for any class- $\mathcal{K}$ function $\tilde{\mu}(\cdot)$, it is iISS with a gain $\mu(\cdot)$ satisfying $\mu(s) = \tilde{\mu}(s)$ for all $s \in [0, 1]$.*

It is proved in [10, Proposition 7] that, for a GAS system $\dot{z} = q(z)$, the solutions $z(t)$ satisfy

$$|z(t)| \leq \alpha(e^{-kt} \gamma(|z(0)|)), \quad (8)$$

where $k > 0$, and $\alpha(\cdot)$ and $\gamma(\cdot)$ are class- $\mathcal{K}_\infty$ functions. The following definition classifies GAS systems using the function $\alpha(\cdot)$ as an index of their speed of convergence to zero.

Definition 3 *Given a class- $\mathcal{K}_\infty$ function $\alpha(\cdot)$, we say that the system $\dot{z} = q(z)$ is GAS(α) if there exists a class- $\mathcal{K}_\infty$ function $\gamma(\cdot)$ and a positive constant $k > 0$ such that (8) holds for all $z(0)$.*

Thus, for the identity function $\alpha(\cdot) = I(\cdot)$, GAS(I) consists of systems in which the convergence is exponential. We next show that $\alpha(\cdot)$ is determined by the local speed of convergence:

Lemma 4 *If the equilibrium $z = 0$ of $\dot{z} = q(z)$ is GAS, and if there exist a constant $\epsilon > 0$, and a $\mathcal{K}_\infty$ function $\tilde{\gamma}(\cdot)$, such that*

$$|z(0)| \leq \epsilon \Rightarrow |z(t)| \leq \alpha(e^{-kt}\tilde{\gamma}(|z(0)|)), \quad (9)$$

then there exists a class- $\mathcal{K}_\infty$ function $\gamma(s) = \mathcal{O}(\tilde{\gamma}(s))$ near $s = 0$, such that (8) holds for all $z(0)$; that is, $\dot{z} = q(z)$ is GAS(α).

The following definition will be used in our cascade result to characterize the admissible iISS gains from the speed of convergence $\alpha(\cdot)$ of the GAS(α) driving subsystem:

Definition 4 *Given a class- $\mathcal{K}$ $\alpha(\cdot)$, we say that the function $\mu(\cdot)$ is class- $\mathcal{H}_\alpha$ if it is class- $\mathcal{K}$ and satisfies*

$$\int_0^1 \frac{(\mu \circ \alpha)(s)}{s} ds < \infty. \quad (10)$$

In particular, for the identity function $\alpha(\cdot) = I(\cdot)$, class- $\mathcal{H}_I$ is defined by

$$\int_0^1 \frac{\mu(s)}{s} ds < \infty. \quad (11)$$

Thus, $\mu(s)$ is class- $\mathcal{H}_I$ if it is class- $\mathcal{K}_o$, or if $\mu(s) \leq s^p$ for some $p > 0$, such as $\mu(s) = \sqrt{s}$.

3 Main Results

We consider the cascade

$$\dot{x} = f(x, z) \quad (12)$$

$$\dot{z} = q(z) \quad (13)$$

where $x \in \mathbb{R}^{n_x}$, $z \in \mathbb{R}^{n_z}$, $f(\cdot, \cdot)$ and $q(\cdot)$ are locally Lipschitz and satisfy $f(0, 0) = 0$, $q(0) = 0$. The stability properties to be analyzed are with respect to the origin $(x, z) = (0, 0)$, which is an equilibrium for (12)-(13).

Our main stability result characterizes the admissible iISS gains for the x -subsystem from the speed of convergence of the z -subsystem.

Theorem 1 *If the z -subsystem (13) is GAS(α) as in (8), and the x -subsystem with input z is iISS with a class- $\mathcal{H}_\alpha$ iISS gain $\mu(\cdot)$ as in (10), then the cascade (12)-(13) is GAS.*

Proof: We note from (8) that

$$\int_0^\infty \mu(|z(\tau)|) d\tau \leq \int_0^\infty (\mu \circ \alpha)(\gamma(|z(0)|)e^{-k\tau}) d\tau$$

$$= \frac{1}{k} \int_0^{\gamma(|z(0)|)} \frac{(\mu \circ \alpha)(s)}{s} ds, \quad (14)$$

where $s := \gamma(|z(0)|)e^{-k\tau}$. From (10),

$$\lambda(s') := \frac{1}{k} \int_0^{s'} \frac{(\mu \circ \alpha)(s)}{s} ds \quad (15)$$

exists for all $s' \geq 0$, and it is class- $\mathcal{K}$ because $\lambda(0) = 0$ and $\frac{(\mu \circ \alpha)(s)}{s} > 0$ for all $s > 0$. Thus, from (4),

$$\alpha(|x(t)|) \leq \beta(|x(0)|, 0) + \lambda(\gamma(|z(0)|)) \quad (16)$$

which proves stability of the cascade (12)-(13). Because $\int_0^\infty \mu(|z(\tau)|) d\tau$ is bounded, (4) implies $x(t) \rightarrow 0$ as $t \rightarrow \infty$, as proved in [10, Proposition 6]. $\square$

If the x -subsystem with input z is ISS as in [9], then from Lemma 3, it is also iISS with a class- $\mathcal{H}_\alpha$ gain. This means that, no matter what the speed of convergence $\alpha(\cdot)$ is for the z -subsystem, the ISS x -subsystem satisfies the corresponding $\mathcal{H}_\alpha$ iISS gain condition of Theorem 1. Thus, Theorem 1 encompasses the following well-known result:

Corollary 1 *If the z -subsystem (13) is GAS and the x -subsystem is ISS, then the cascade (12)-(13) is GAS.* $\square$

Another particular case of interest is when the z -subsystem is LES, that is when (9) holds with $\alpha(\cdot) = I(\cdot)$ and $\tilde{\gamma}(s) = cs$ for some $c \geq 1$. From Lemma 4, there exist a class- $\mathcal{K}_o$ function $\gamma(\cdot)$, and a constant $k > 0$ such that, for all $z(0)$,

$$|z(t)| \leq e^{-kt}\gamma(|z(0)|). \quad (17)$$

This means that the z -subsystem is GAS(I) and, hence, Theorem 1 requires that the iISS gain be class- $\mathcal{H}_I$:

Corollary 2 *If the z -subsystem (13) is GAS and LES, and the x -subsystem is iISS with a class- $\mathcal{H}_I$ gain $\mu(\cdot)$ as in (11), then the cascade (12)-(13) is GAS.* $\square$

We note from Lemma 2 that the class- $\mathcal{H}_I$ restriction of Corollary 2 is satisfied when the iISS x -subsystem is affine in z :

Corollary 3 *If the z -subsystem (13) is GAS and LES, and the x -subsystem is iISS and affine in z , then the cascade (12)-(13) is GAS.* $\square$

Examples 1 and 2 below illustrate that the LES and the class- $\mathcal{H}_I$ gain restrictions cannot be removed from Corollary 2:

Example 1 For the cascade system

$$\dot{x} = -\text{sat}(x) + xz \quad (18)$$

$$\dot{z} = -z^3, \quad (19)$$

where $\text{sat}(x) := \text{sgn}(x) \min\{1, |x|\}$, the x -subsystem with input z is iISS with a class- $\mathcal{K}_o$ (hence, class- $\mathcal{H}_I$) gain, as verified from $V(x) = \frac{1}{2} \ln(1 + x^2)$ which satisfies $\dot{V} \leq |z|$ as in Lemma 1. However, the cascade (18)-(19) has unbounded solutions because the z -subsystem is not LES. To prove this, we let $z(0) = 1$, so that $z(t) = \frac{1}{\sqrt{1+2t}}$, and $x(0) > 1$, so that as long as $x(t) \geq 1$,

$$\dot{x} = \frac{1}{\sqrt{1+2t}} x - 1 \Rightarrow \quad (20)$$

$$x(t) = e^{(\sqrt{1+2t}-1)} \left[x(0) - \int_0^t e^{(1-\sqrt{1+2\tau})} d\tau \right].$$

Using the change of variables $s = -1 + \sqrt{1+2\tau}$, we obtain

$$\begin{aligned} \int_0^t e^{(1-\sqrt{1+2\tau})} d\tau &\leq \int_0^\infty e^{(1-\sqrt{1+2\tau})} d\tau \quad (21) \\ &= \int_0^\infty e^{-s} (s+1) ds = 2, \end{aligned}$$

thus, if $x(0) \geq 3$, then (20) implies $x(t) \geq e^{(\sqrt{1+2t}-1)}$. This means that $x(t) \geq 1$ for all $t \geq 0$ and $x(t) \rightarrow \infty$ as $t \rightarrow \infty$. $\square$

Example 2 Consider the locally Lipschitz system

$$\begin{aligned} \dot{x} &= -m(x) + g^{-1} \left(g \left(e^{\frac{x-1}{b}} - a z^2 \right) \text{sat}(z^2)/2 \right) \\ \dot{z} &= -z/2 \end{aligned} \quad (22)$$

where $g : [0, +\infty) \rightarrow [0, 1]$ is defined as $g(x) := e^{-\frac{x}{b}}$ for $x > 0$ and $g(0) = 0$, $m(x)$ is a locally Lipschitz function satisfying $m(x)x > 0$ for all $x \neq 0$, and

$$m(x) = \frac{1}{2e^{\frac{|x|-1}{b}}} \quad \forall x \geq 1 + b \ln(a), \quad (23)$$

and $a, b > 0$ are constants to be specified. Using $V(x) = \ln(1 + x^2)$, and $|g(r)| \leq 1$ for all $r \in \mathbb{R}_{\geq 0}$, we obtain

$$\begin{aligned} \dot{V} &= -\frac{2m(x)x}{1+x^2} \\ &\quad + \frac{2x}{1+x^2} g^{-1} \left(g \left(e^{\frac{x-1}{b}} - a z^2 \right) \text{sat}(z^2)/2 \right) \\ &\leq -\frac{2m(x)x}{1+x^2} + g^{-1}(\text{sat}(z^2)/2), \end{aligned} \quad (24)$$

which, from Lemma 1, proves that the x -subsystem is iISS with input z . Moreover, the z -subsystem is exponentially stable as in Corollary 2. However, because the iISS gain is not class- $\mathcal{H}_I$, the cascade (22) admits the unbounded solution

$$x(t) = 1 + b \ln(a + t) \quad (25)$$

when $z(0) \in (0, 1)$, $a = \ln(2/z^2(0)g(z^2(0)))$, and $b = 1/2$. $\square$

Theorem 1 characterized the class of admissible iISS gains from the speed of convergence of the z -subsystem. It may appear that this class can be enlarged by a change of coordinates in which the z -subsystem converges faster as in Grüne *et al.* [1]. The following example illustrates that such an attempt fails, because in the new coordinates the iISS gain of the x -subsystem becomes steeper:

Example 3 For the system (18)-(19), the change of coordinates

$$\tilde{z} = \Phi(z) := z e^{-\frac{1}{2z^2}} \quad (26)$$

ensures exponential convergence for the $\tilde{z}$ -subsystem:

$$\dot{\tilde{z}} = -\text{sat}(\tilde{z}) + \tilde{z} \Phi^{-1}(\tilde{z}) \quad (27)$$

$$\dot{\tilde{z}} = -(1 + [\Phi^{-1}(z)]^2) \tilde{z}. \quad (28)$$

Using $V(x) = \frac{1}{2} \ln(1 + x^2)$, we obtain

$$\dot{V} \leq -\frac{x \text{sat}(x)}{1+x^2} + \Phi^{-1}(|\tilde{z}|),$$

which, from Lemma 1, implies that the x -subsystem with input $\tilde{z}$ is iISS with gain $\Phi^{-1}(\cdot)$. The cascade (27)-(28) has unbounded solutions as proved in Example 1, because all derivatives of $\Phi(z)$ vanish at $z = 0$ and, hence, the inverse function $\Phi^{-1}(\cdot)$ is too steep at zero to satisfy the class- $\mathcal{H}_I$ condition of Corollary 2. $\square$

4 Growth Restrictions on the Interconnection Term

It is well known that the nonlinear growth of the interconnection term $h(x, z) := f(x, z) - f(x, 0)$ plays an important role for the stability of the cascade (12)-(13), rewritten here as

$$\dot{x} = f(x, 0) + h(x, z) \quad (29)$$

$$\dot{z} = q(z). \quad (30)$$

Indeed, most cascade results restrict the growth of $h(x, z)$ in x to be linear; see *e.g.* Janković *et al.* [2], Sepulchre *et al.* [7, Chapter 4], and Panteley and Loria [5]. In this section we show that the class- $\mathcal{H}_I$ gain condition of Corollary 2 imposes a similar growth restriction, which, however, is less restrictive and allows the growth of $h(x, z)$ in x to be faster than linear.

We first show that Corollary 2 encompasses the results in which the growth of $h(x, z)$ in x is linear:

Proposition 1 (Janković *et al.* [2]) *Suppose that the system (29)-(30), where $\dot{x} = f(x, 0)$ is GAS and $\dot{z} = q(z)$ is GAS and LES, satisfies the following assumptions:*

A1) *The interconnection term $h(x, z)$ has linear growth in x , that is,*

$$|h(x, z)| \leq \gamma_1(|z|) + \gamma_2(|z|)|x| \quad (31)$$

for some class- $\mathcal{K}_o$ functions $\gamma_1(\cdot)$ and $\gamma_2(\cdot)$.

A2) *There exist positive constants r and k such that the Lyapunov function $U(x)$, which establishes global stability of $\dot{x} = f(x, 0)$ with $\frac{\partial U}{\partial x} f(x, 0) \leq 0$, also satisfies*

$$|x| \geq r \Rightarrow \left| \frac{\partial U}{\partial x} \right| |x| \leq kU(x). \quad (32)$$

Then, the cascade (29)-(30) is GAS. $\square$

Proof: To show that the result follows from Corollary 2, we prove that A1 and A2 imply that the x -subsystem is iISS with a class- $\mathcal{H}_I$ gain $\mu(\cdot)$. To this end, we define

$$V(x) = \begin{cases} \ln(U(x)) & \text{if } U(x) > e \\ \frac{1}{e}U(x) & \text{if } U(x) \leq e, \end{cases} \quad (33)$$

which is C^1 , positive definite, and radially unbounded, and assume that $U(x) \geq e \Rightarrow |x| \geq r$. (If this is not the case, scale $U(x)$ by a sufficiently small constant c so that $\tilde{U}(x) := cU(x) \geq e \Rightarrow |x| \geq r$, and note that $\tilde{U}(x)$ also satisfies A2.) Thus,

$$\begin{aligned} U(x) \geq e &\Rightarrow \\ \frac{\partial V}{\partial x}[f(x, 0) + h(x, z)] &\leq \frac{1}{U} \frac{\partial U}{\partial x} h(x, z) \\ &\leq \frac{1}{U} \left| \frac{\partial U}{\partial x} \right| |h(x, z)|. \end{aligned} \quad (34)$$

Noting from (31) that $|x| \geq r$ implies $|h(x, z)| \leq \left(\frac{1}{r}\gamma_1(|z|) + \gamma_2(|z|)\right)|x|$, and using (34) and (32), we obtain

$$\begin{aligned} U(x) \geq e &\Rightarrow \\ \frac{\partial V}{\partial x}[f(x, 0) + h(x, z)] &\leq k \left(\frac{1}{r}\gamma_1(|z|) + \gamma_2(|z|) \right) \\ &=: \gamma_3(|z|). \end{aligned} \quad (35)$$

Likewise, because $\left| \frac{\partial U}{\partial x} \right|$ is bounded on compact sets, and the function $h(x, z)$ is locally Lipschitz, and satisfies $h(x, 0) = 0$ for all x , we can find a class- $\mathcal{K}_o$ function $\gamma_4(\cdot)$, such that

$$\begin{aligned} U(x) \leq e &\Rightarrow \\ \frac{\partial V}{\partial x}[f(x, 0) + h(x, z)] &\leq \frac{1}{e} \left| \frac{\partial U}{\partial x} \right| |h(x, z)| \\ &\leq \gamma_4(|z|). \end{aligned} \quad (36)$$

From (34) and (36), $V(x)$ satisfies $\dot{V} \leq \sigma(|z|)$ with the class- $\mathcal{K}_o$ function $\sigma(\cdot) = \max\{\gamma_3(\cdot), \gamma_4(\cdot)\}$. Thus, from Lemma 1, the x -subsystem with input z is iISS with a class- $\mathcal{K}_o$ (hence, class- $\mathcal{H}_I$) gain $\mu(\cdot)$. $\square$

We wish to emphasize that Corollary 2 allows the growth of $h(x, z)$ in x to be faster than linear. To illustrate this, we consider the cascade (29)-(30) with $x \in \mathbb{R}$, and $h(x, z)$ restricted by

$$|h(x, z)| \leq \gamma_1(|z|) + \gamma_2(|z|)\varphi(|x|), \quad (37)$$

which is (31) when $\varphi(|x|) = |x|$. We characterize the class- $\mathcal{K}$ functions $\varphi(\cdot)$ for which the x -subsystem satisfies the class- $\mathcal{H}_I$ gain condition of Corollary 2:

Proposition 2 *Consider the cascade (29)-(30), where $x \in \mathbb{R}$. If $\dot{x} = f(x, 0)$ is GAS, $\dot{z} = q(z)$ is GAS and LES, and $h(x, z)$ satisfies (37) for some class- $\mathcal{H}_I$ functions $\gamma_1(\cdot)$, $\gamma_2(\cdot)$, and a class- $\mathcal{K}$ function $\varphi(\cdot)$ satisfying*

$$\int_1^\infty \frac{1}{\varphi(s)} ds = \infty, \quad (38)$$

then the origin is GAS. $\square$

Proof: To prove that the x -subsystem is iISS with a class- $\mathcal{H}_I$ gain, we let $V(x)$ be a smooth, positive definite function such that $V(x) = V(-x)$, and

$$x \geq 1 \Rightarrow V(x) = V(1) + \int_1^x \frac{1}{\varphi(s)} ds. \quad (39)$$

Because of (38), $V(x)$ is radially unbounded, and satisfies

$$|x| \geq 1 \quad \Rightarrow \quad \left| \frac{\partial V}{\partial x} \right| = \frac{1}{\varphi(|x|)}. \quad (40)$$

Thus, if $|x| \geq 1$, (37) and (40) yield

$$\begin{aligned} \frac{\partial V}{\partial x}[f(x, 0) + h(x, z)] &\leq \frac{1}{\varphi(1)}\gamma_1(|z|) + \gamma_2(|z|) \\ &=: \gamma_5(|z|). \end{aligned} \quad (41)$$

If $|x| \leq 1$, then $\left| \frac{\partial V}{\partial x} \right| \leq b$ for some positive constant b , and

$$\begin{aligned} \frac{\partial V}{\partial x}[f(x, 0) + h(x, z)] &\leq b\gamma_1(|z|) + b\gamma_2(|z|)\varphi(1) \\ &:= \gamma_6(|z|). \end{aligned} \quad (42)$$

Because $\dot{V} \leq \max\{\gamma_5(|z|), \gamma_6(|z|)\}$, the x -subsystem is iISS with a class- $\mathcal{K}_I$ gain as in Corollary 2 and, hence, the cascade (29)-(30) is GAS. $\square$

It is important to note that the growth condition (38) encompasses functions that grow faster than linear, such as $\varphi(|x|) = |x| \ln(|x|)$. On the other hand, (38) disallows $\varphi(|x|) = |x|^2$, $\varphi(|x|) = |x|^3$, etc. We finally show that (38) is tight, and cannot be relaxed:

Example 4 The cascade

$$\dot{x} = -\text{sat}(x) + \varphi(x)z \quad (43)$$

$$\dot{z} = -z, \quad (44)$$

exhibits finite escape time when the class- $\mathcal{K}$ function $\varphi(x)$ fails to satisfy (38), that is,

$$\int_1^\infty \frac{1}{\varphi(s)} ds < \infty. \quad (45)$$

To prove this, we let $V(x)$ be as in (39), and note from (45) that there exists a constant $V_\infty > 0$ such that $V(x) < V_\infty$ for all $x \in \mathbb{R}$, and $V(x) \rightarrow V_\infty$ as $x \rightarrow \infty$. Moreover, from (39),

$$\begin{aligned} x(t) \geq 1 &\Rightarrow \\ \dot{V} &= -\frac{1}{\varphi(x(t))} + z(t) \geq -\frac{1}{\varphi(1)} + z(0)e^{-t}. \end{aligned} \quad (46)$$

If $z(0) > 1/\varphi(1)$, then we can find $T > 0$ such that,

$$\begin{aligned} \phi(t) &:= \int_0^t \left(-\frac{1}{\varphi(1)} + z(0)e^{-t} \right) dt \\ &= -\frac{t}{\varphi(1)} + z(0)(1 - e^{-t}) \end{aligned} \quad (47)$$

satisfies $\phi(t) > 0$ for all $t \in (0, T]$. Thus, if $x(0)$ is such that $x(0) \geq 1$ and $V_\infty - \phi(T) \leq V(x(0)) < V_\infty$, then it follows from (46) that

$$V(x(T)) \geq V(x(0)) + \phi(T) \geq V_\infty,$$

which proves that $x(T)$ is not defined. $\square$

5 Conclusion

We have studied stability of nonlinear cascades, and showed that a trade-off exists between slower convergence for the driving subsystem and steeper iISS gain for the driven subsystem. This approach unifies several results in the literature, obtained by restricting the speed of convergence of the driving subsystem, and the nonlinear growth of the driven subsystem. We have studied the connection between these growth conditions and the iISS gain, and proved that our iISS gain restriction leads to a less restrictive condition than the linear growth assumption common in the literature. It would be of interest to extend our cascade result to feedback interconnections, where small-gain formulations of iISS can be pursued.

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